School of Mathematical Sciences

Autumn Semester 2025-2026

MATH4065- STATISTICAL FOUNDATIONS

Assessed Coursework

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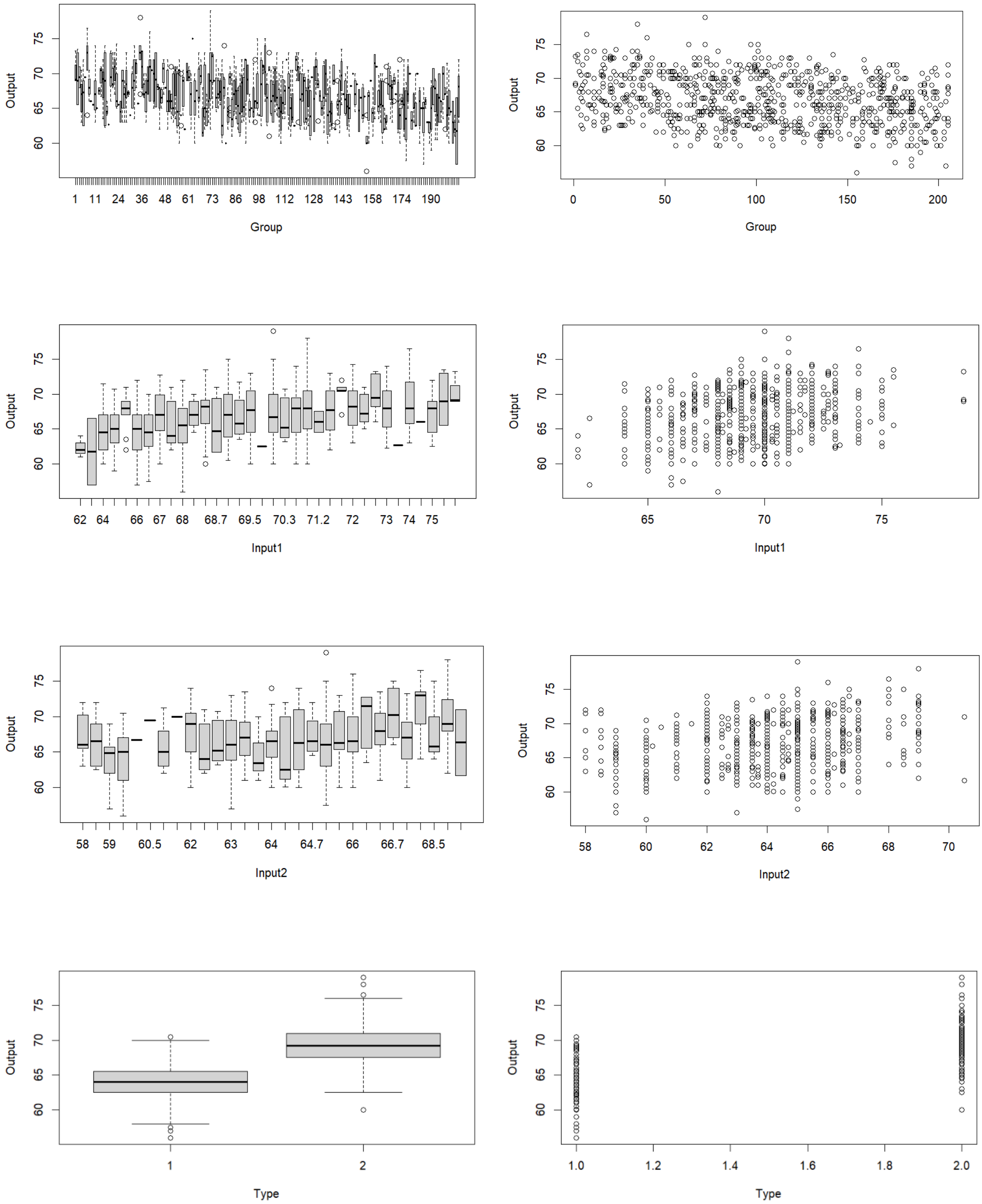
Exercise 1

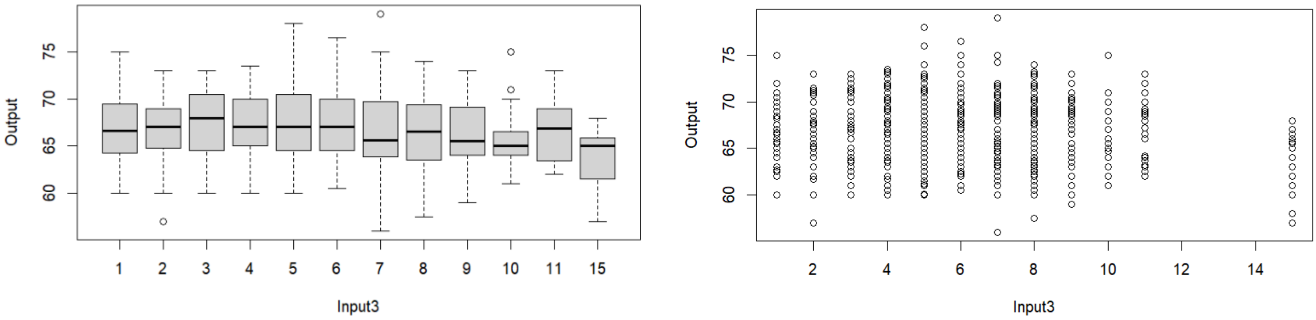
(a) In order to check whether there is any graphical evidence of relationships between Output and the other five variables we will use the following R commands to plot each predictor variable against the response variable Output in both boxplots and scatter plots.

The code used to generate the plots

```
boxplot(Output~Group)
plot(Group, Output)

boxplot(Output~Input1)
plot(Input1, Output)

boxplot(Output~Input2)
plot(Input2, Output)

boxplot(Output~Type)
plot(Type, Output)

boxplot(Output~Input3)
plot(Input3, Output)
```

The plots generated:





At first glance it does not look like there is a one-to-one relationship between the input variables and the output variable. Normally we expect the output variable to have a relationship with a combination of input variables not just one-to-one without having into account the rest of the input variables.

(b) With this R code we will fit the linear model in which Output is the response variable and the five other variables are all included as predictor variables as well as a constant term:

```
out<-lm(Output~Group + Input1 + Input2 + Type + Input3, data=data)
summary(out)
```

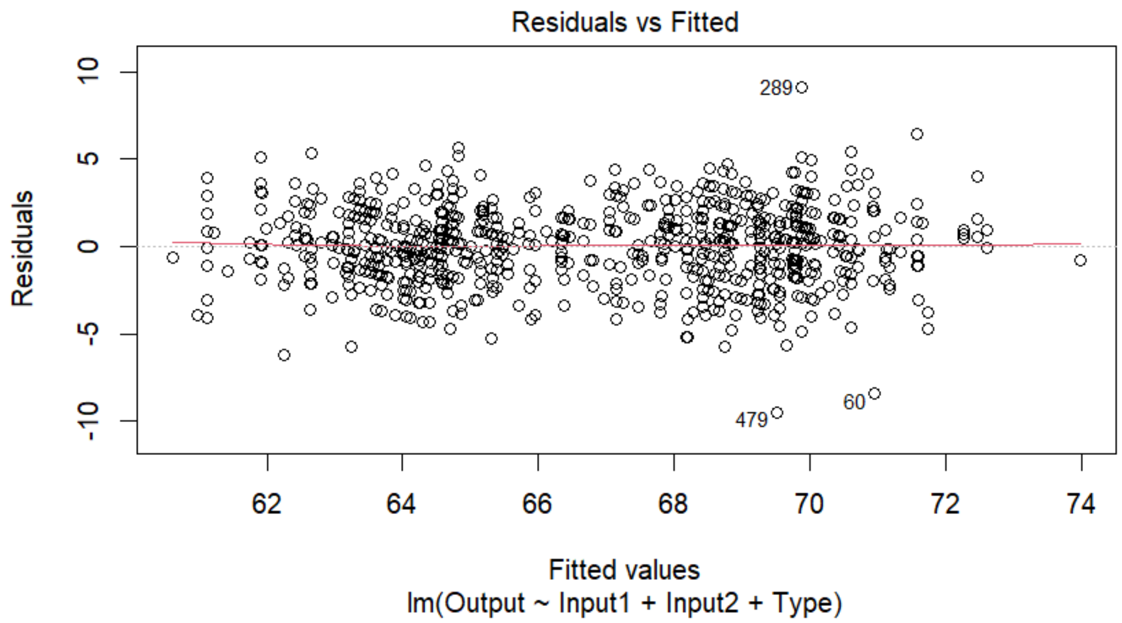
This code provided the following output:

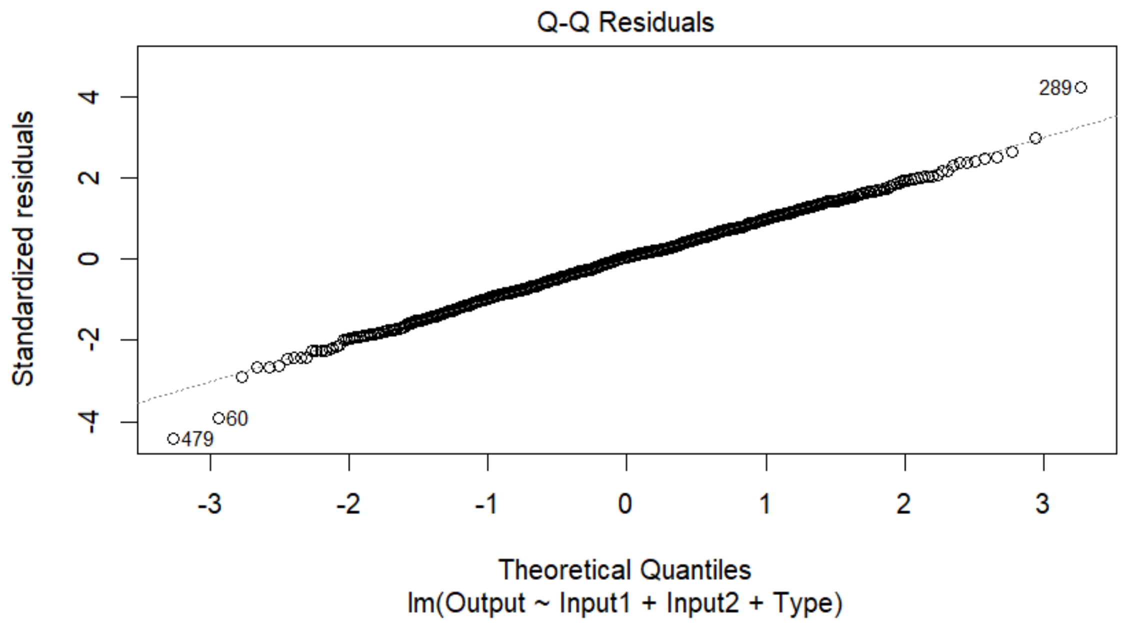
```
Call:
lm(formula = Output ~ Group + Input1 + Input2 + Type + Input3,
    data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-9.4152 -1.4036  0.1048  1.4398  9.0721

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 21.891208   8.030314   2.726  0.00653 **
Group       -0.006162   0.004248  -1.451  0.14725
Input1       0.264727   0.096711   2.737  0.00632 **
Input2       0.304585   0.033212   9.171  < 2e-16 ***
Type         5.216492   0.144203  36.175  < 2e-16 ***
Input3      -0.040222   0.027275  -1.475  0.14065
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.151 on 892 degrees of freedom
Multiple R-squared:  0.6416,    Adjusted R-squared:  0.6396
F-statistic: 319.3 on 5 and 892 DF,  p-value: < 2.2e-16
```

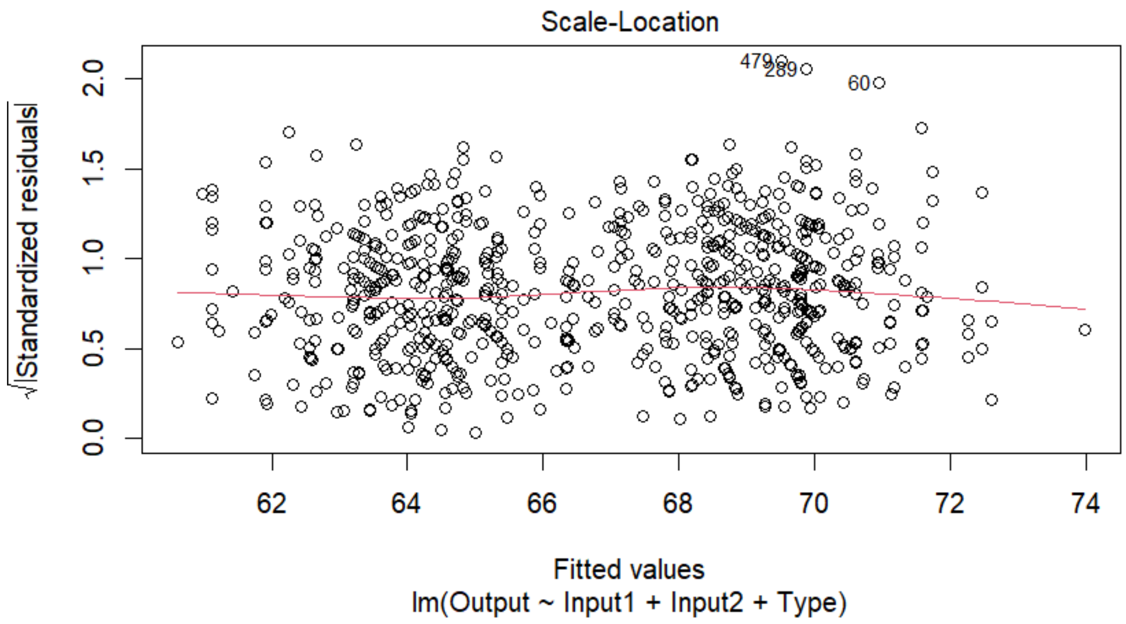
Where the estimate column is indeed the least-square estimates of the model parameters. Notice that the p-values of Group (0.14725) and Input3 (0.14065) are greater than 0.10 which, according to the class notes, means that they provide no evidence to reject H_0 which means that those predictors do not provide predictive information when the others are already included. Our new linear model with these variables removed can be computed with the following code:

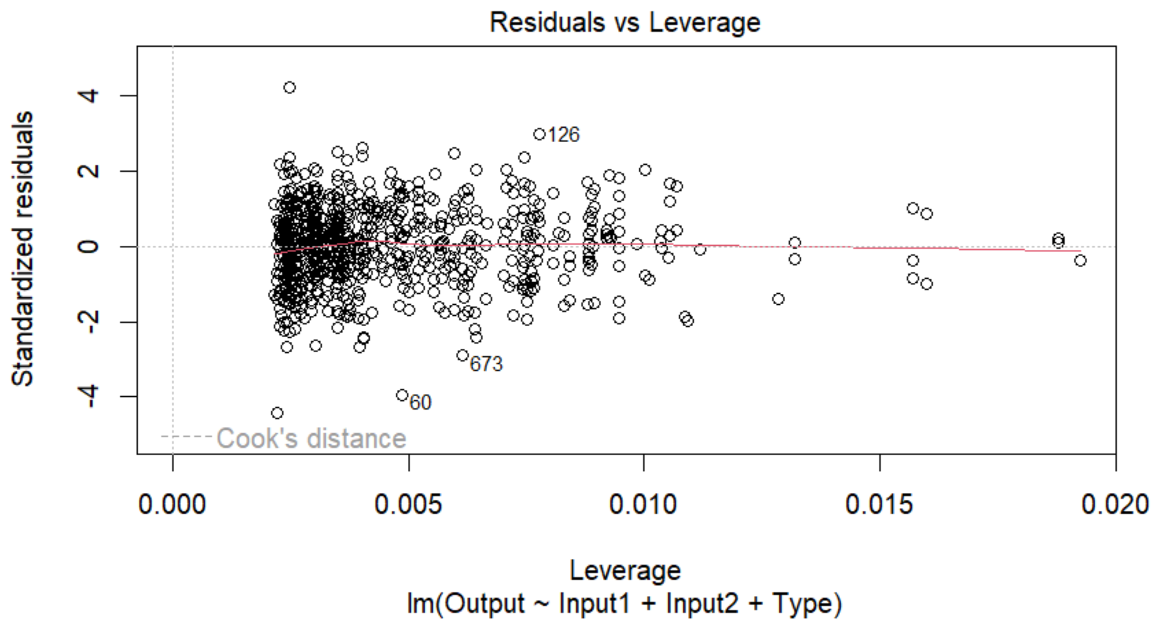
```
out<-lm(Output~ Input1 + Input2 + Type, data=data)
summary(out)
plot(out)
```

(c) This code generates the following images:









The Q-Q Residuals graph compares the standardized residuals against the theoretical quantiles. As the points approach the theory line along the distribution, we can say that the residuals follow a normal distribution.

Exercise 2

(a) The probability density function given corresponds to the Weibull distribution:

$$f(x) = \left(\frac{\theta}{\lambda}\right)\left(\frac{x}{\lambda}\right)^{(\theta-1)} e^{-\left(\frac{x}{\lambda}\right)^{\theta}}, \qquad x > 0, \theta > 0, \lambda = 1$$

The log-likelihood function is defined as $l(\theta) = log\, L(\theta)$ where $L(\theta) = \prod_{i=1}^{n} f(x_i|\theta)$ the likelihood function for a random sample.

$$L(\theta) = \prod_{i=1}^{n} f(x_i|\theta) = \prod_{i=1}^{n} \theta {x_i}^{\theta-1} e^{-{x_i}^{\theta}} = \theta^n \prod_{i=1}^{n} \left(x_i^{\theta-1}\right) e^{-\sum_{i=1}^{n} {x_i}^{\theta}}$$

Taking logarithms in both sides, we get:

$$l(\theta) = log\left(\theta^n \prod_{i=1}^{n} \left(x_i^{\theta-1}\right) e^{-\sum_{i=1}^{n} x_i^{\theta}}\right) = log(\theta^n) + log\left(\prod_{i=1}^{n} x_i^{\theta-1}\right) + log\left(e^{-\sum_{i=1}^{n} x_i^{\theta}}\right)$$

$$= n \log\theta + \sum_{i=1}^{n} log\left(x_i^{\theta-1}\right) - \sum_{i=1}^{n} x_i^{\theta} = n\, log\,\theta + (\theta - 1)\sum_{i=1}^{n} log(x_i) - \sum_{i=1}^{n} x_i^{\theta}$$

(b) The following R code finds the maximum likelihood estimate of θ given the observed data provided by the question:

```
dataset<-c(0.610, 0.309, 0.578, 0.921, 1.04, 0.820, 1.71, 0.834,
0.852, 1.11)

log.lik.weibull<-function(theta,x){
  if (theta <= 0){
    return(-Inf)
  }
  n<-length(x)
  sum_l = sum(log(x))
  result<-n*log(theta)+(theta-1)*sum_l-sum(x^theta)
  return(result)
}

minus.log.lik.weibull<-function(theta,x){
  -log.lik.weibull(theta,x)
}
t<-optimise(minus.log.lik.weibull, c(0,10), x = dataset)
print(t)
```

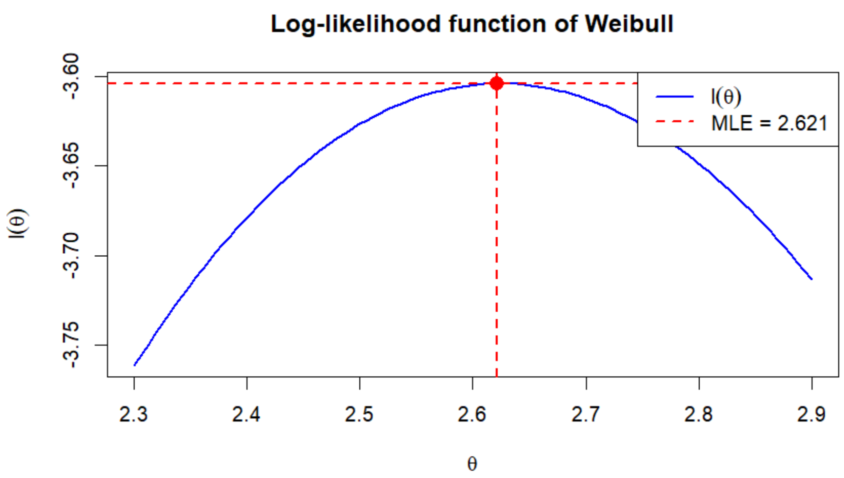
This code provides the following output:

```
$minimum
[1] 2.623884

$objective
[1] 3.604138
```

Which implies $\hat{\theta} = 2.623884$ and the value of the objective function (i.e. log-likelihood) at $\hat{\theta}$ is 3.604138.

(c) The following code produces a plot of $l(\theta)$ over a suitable range of values with the maximum likelihood estimate of θ shown. The comments explain the logic behind the code.

```
#Generate theta values
theta.values<-seq(2.3,2.9, len=100)

# Compute the log-likelihood function for each theta
log_lik_values<-vector(length = length(theta.values))
for (i in 1:length(theta.values)){
  log_lik_values[i]<- log.lik.weibull(c(theta.values[i]), dataset)
}

# Find MLE in data generated
mle_theta <- theta.values[which.max(log_lik_values)]

# Plot log-likelihood
plot(theta.values, log_lik_values,
     type = "l",
     lwd = 2,
     col = "blue",
     xlab = expression(theta),
     ylab = expression(l(theta)),
     main = "Log-likelihood function of Weibull")

# Show MLE in graphic
abline(h= -t[["objective"]], v = mle_theta, col = "red", lwd = 2, lty = 2)
points(mle_theta, max(log_lik_values), col = "red", pch = 16, cex = 1.5)

# Add legend
legend("topright",
       legend = c(expression(l(theta)), paste("MLE =", round(mle_theta, 3))),
       col = c("blue", "red"), lwd = c(2, 2), lty = c(1, 2))
```



This plot shows how the MLE we found maximizes the log-likelihood function.

Exercise 3

The following R code finds

(a) $E\left[\sqrt{\left|\frac{X}{Y}\right|}\right]$

(b) $\boldsymbol{P(sin(XY) > 0.5)}$

(c) $\boldsymbol{P(X < Z)}$

```
n<-100000
x<-rnorm(n)
y<-rnorm(n,mean = 1, sd = sqrt(2))
z<-rexp(n, rate = abs(x))
#(a)
mean(sqrt(abs(x/y)))
#(b)
mean(sin(x*y) > 0.5) #P(sin(XY)>0.5)
#(c)
mean(x<z)
```

The results are:

(a) $\mathbf{1.053983}$
(b) $\mathbf{0.24455}$
(c) $\mathbf{0.7898}$